

# **Software Requirements Specification (SRS)**

## **Food Donation and Lifecycle Management System**

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**SE Lab 3B - Group C**

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## 1. Introduction

### 1.1 Purpose of the System

The purpose of this system is to specify and model a Food Donation and Lifecycle Management platform that connects food donors with NGOs, volunteers, delivery partners, and food safety inspectors to reduce food wastage and ensure safe, efficient food redistribution.

The system aims to:

1. Allow food donors to register, list, edit, and cancel food donations
2. Enable NGOs to view, accept, or reject donations based on food safety inspections
3. Coordinate volunteers and delivery partners to manage pickup and delivery logistics
4. Enforce food safety standards through a dedicated inspector role
5. Provide system administrators with tools to manage users, monitor activity, and generate reports
6. Track the complete donation lifecycle from submission to delivery

### 1.2 Scope of the System

#### 1.2.1 The system will

1. Allow food donors to register and submit food donation requests

2. Enable NGOs to accept or reject donations after food safety inspection
3. Support volunteer management including pickup task assignment and status updates
4. Allow delivery partners to accept, pick up, and deliver food donations
5. Provide food safety inspectors with tools to inspect donations, define rules, and flag unsafe food
6. Give system administrators control over user management, records, notifications, and reporting
7. Track the full donation lifecycle in real time
8. Automate donation matching between donors and NGOs
9. Support a feedback and rating system for completed donations

### **1.2.2 The system will not**

1. Handle financial transactions or monetary compensation
2. Connect to external government food regulatory databases
3. Replace physical food inspection processes
4. Manage restaurant or commercial food supply chains

## **1.3 Objectives and Success Criteria**

### **1.3.1 Objectives**

1. Minimize food wastage through efficient donation matching
2. Ensure all donated food meets safety standards before distribution
3. Streamline volunteer and delivery coordination
4. Maintain complete and traceable donation lifecycle records
5. Enforce strict role-based access control across all modules

### **1.3.2 Success Criteria**

The system is successful if:

1. Donations follow valid lifecycle transitions from submission to delivery
2. Food safety inspections are completed before acceptance or rejection
3. Volunteers and delivery partners receive real-time task notifications
4. NGOs can match donations to beneficiaries efficiently
5. Administrators can generate accurate reports on donation activity
6. Unauthorized users cannot access restricted modules

## **1.4 Definitions, Acronyms and Abbreviations**

1. Donation Lifecycle — Sequential stages from submission to delivery
2. Reconciliation — Matching of donation records with delivery confirmations

3. Break — A detected mismatch or issue in donation records
4. NGO — Non-Governmental Organization acting as a food bank
5. Pickup Task — Assignment given to a volunteer for collecting a donation
6. Delivery Task — Assignment given to a delivery partner for delivering food
7. Flag — Marking a food donation as unsafe for distribution
8. RBAC — Role-Based Access Control

## 1.5 References

1. Food Donation Lifecycle — Overview of food redistribution workflows
2. Food Safety Standards — FSSAI Guidelines on Food Handling
3. Volunteer Coordination in Non-Profit Systems
4. IEEE Requirements Engineering Standard (ISO/IEC/IEEE 29148)

## 1.6 Overview

This document specifies the functional and non-functional requirements of the proposed Food Donation and Lifecycle Management System including all actors, use cases, scenarios, and system constraints.

# 2. Current System

Traditional food donation and redistribution systems:

1. Rely on manual phone or email coordination between donors and NGOs
2. Lack real-time visibility into donation availability and status
3. Have no formal food safety inspection workflow
4. Depend on physical registers for tracking donations
5. Provide no structured feedback mechanism for donors

Problems include:

1. Food wastage due to coordination delays
  2. Risk of distributing unsafe or expired food
  3. No accountability for volunteers or delivery partners
  4. Difficulty generating reports on donation activity
  5. Lack of donor engagement and trust
-

# 3. Proposed System

## 3.1 Overview

The proposed system introduces a centralized digital platform with role-based access for all stakeholders.

### 3.1.1 Centralized Layer

1. Donation listing and management
2. Lifecycle tracking
3. Volunteer and delivery coordination

### 3.1.2 Management Layer

1. Food safety inspection workflow
2. Donation matching system
3. Real-time notifications and reporting

## 3.2 Functional Requirements

### 3.2.1 User Management

FR1: The system shall implement role-based authentication for all users. FR2: The system shall support the following user roles:

1. Food Donor
2. NGO / Food Bank
3. Volunteer
4. Delivery Partner
5. System Administrator
6. Food Safety Inspector

FR3: The system shall prevent unauthorized users from accessing restricted modules.

### 3.2.2 Donation Management

FR4: The system shall allow food donors to submit, edit, and cancel food donation requests.

FR5: The system shall enforce valid donation lifecycle transitions. The allowed states are:

| Current State | Next Allowed State |
|---------------|--------------------|
|---------------|--------------------|

|                 |                     |
|-----------------|---------------------|
| Submitted       | Inspected           |
| Inspected       | Accepted / Rejected |
| Accepted        | Pickup Assigned     |
| Pickup Assigned | Picked Up           |
| Picked Up       | Delivered           |
| Delivered       | Reconciled          |

FR6: The system shall reject any attempt to skip or revert lifecycle stages.

### **3.2.3 Food Safety**

FR7: The system shall require food safety inspection before any donation is accepted or rejected. FR8: The system shall allow food safety inspectors to flag unsafe donations. FR9: The system shall allow NGOs to manage unsafe food and penalize donors for violations.

### **3.2.4 Volunteer and Delivery Coordination**

FR10: The system shall allow volunteers to view pickup requests, set availability, and accept pickup tasks. FR11: The system shall allow delivery partners to accept delivery tasks, pick up food, and update delivery status. FR12: The system shall allow volunteers to report issues encountered during pickup.

### **3.2.5 NGO Operations**

FR13: The system shall provide a donation matching system that maps available donations to NGO needs. FR14: The system shall allow NGOs to manage volunteers and delivery partners. FR15: The system shall allow NGOs to assign and manage food safety inspectors.

### **3.2.6 Reporting and Administration**

FR16: The system shall allow system administrators to manage all users and records. FR17: The system shall generate reports based on donation lifecycle data. FR18: The system shall provide a real-time notification system for all role-based events.

## **3.3 Non-Functional Requirements**

### **3.3.1 Performance**

NFR1: The system shall support up to 500 concurrent donation listings without performance degradation. NFR2: The system shall deliver real-time notifications within 3 seconds of a trigger event.

### **3.3.2 Security**

NFR3: The system shall store all passwords using salted cryptographic hashes. NFR4: The system shall enforce role-based access control on all modules.

### **3.3.3 Reliability**

NFR5: The system shall maintain ACID-compliant database transactions. NFR6: The system shall reject any lifecycle transition that violates the defined donation lifecycle model.

### **3.3.4 Scalability**

NFR7: The system shall support at least 50 concurrent users within the operational environment without performance degradation.

### **3.3.5 Maintainability**

NFR8: The system shall follow a modular architecture consisting of:

1. UI Layer
2. Business Logic Layer
3. Database Layer
4. Notification Layer

## **3.4 Scenarios**

### **3.4.1 Successful Donation and Delivery**

A food donor submits a donation listing with food type, quantity, and pickup location. The NGO views the available donation and assigns a food safety inspector. The inspector inspects the food and clears it. The NGO accepts the donation and assigns a volunteer for pickup. The volunteer picks up the food and updates the status. A delivery partner then collects it from the volunteer and delivers it to the beneficiary, updating the delivery status upon completion.

### **3.4.2 Unsafe Food Detection**

A food donor submits a donation. The food safety inspector reviews it and finds the food to be contaminated or past expiry. The inspector flags the food as unsafe. The NGO receives the flag, manages the unsafe food by arranging disposal, and issues a penalty to the food donor for submitting non-compliant food. The donation lifecycle is terminated at this stage.

### **3.4.3 Volunteer Pickup Issue**

A volunteer accepts a pickup task and proceeds to the donor's location. During pickup, the volunteer encounters a problem — the food quantity does not match what was listed, or the donor is unavailable. The volunteer reports the issue through the system by extending the Update Pickup Status use case. The NGO is notified and takes corrective action.

### **3.4.4 Donation Matching**

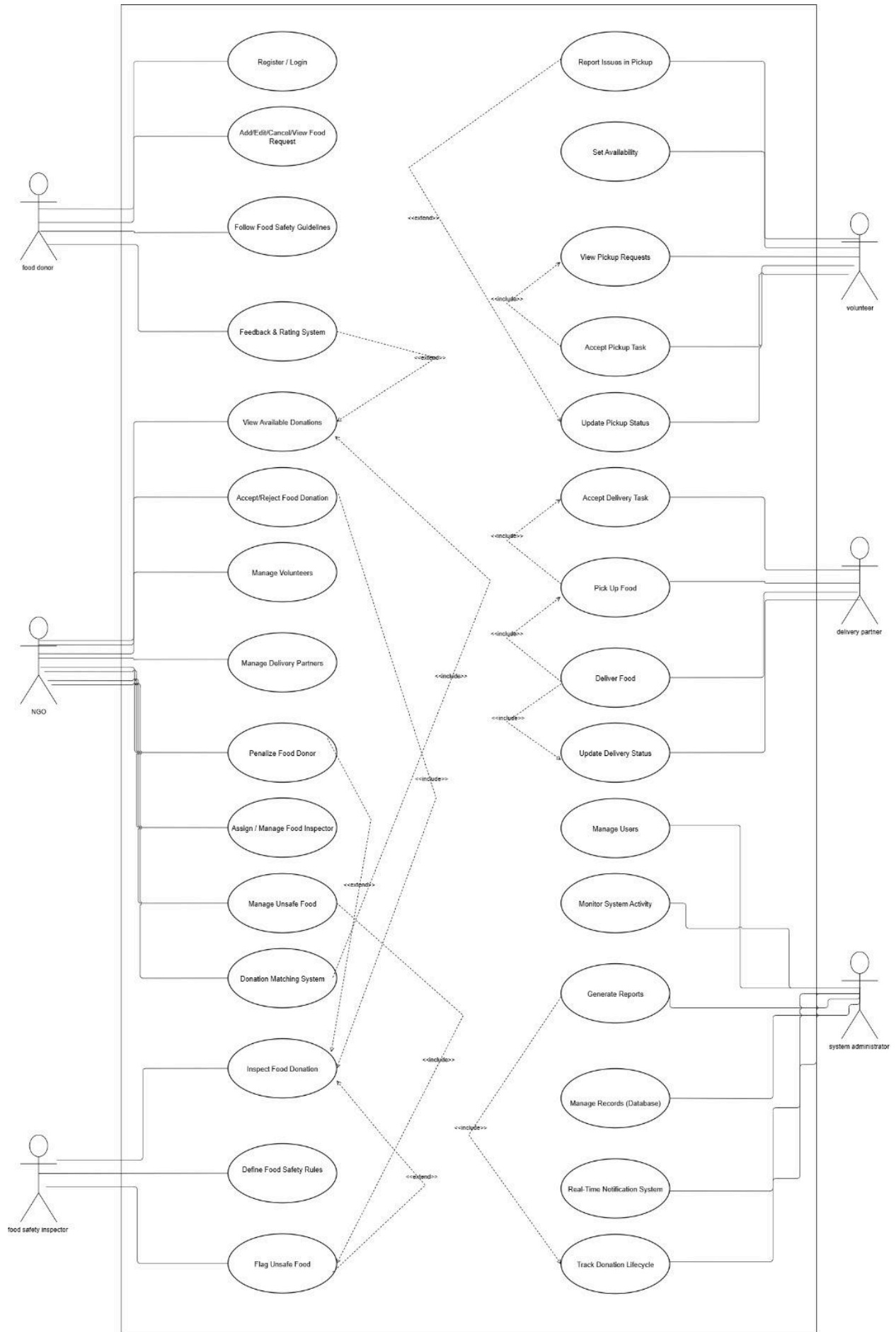
An NGO needs food supplies for a specific beneficiary group. The donation matching system scans all available donations and suggests the best match based on food type, quantity, and proximity. The NGO reviews the suggested match and accepts the most suitable donation, initiating the standard acceptance workflow.

### **3.4.5 Invalid Login Attempt**

A user attempts to log into the system by entering a username and password. The system validates the entered credentials against stored user records. If the credentials are incorrect, the system denies access, displays an error message, and logs the failed attempt for security monitoring. The user may retry with correct credentials.

## **3.5 Use Case Model**





**Actors:**

1. Food Donor
2. NGO / Food Bank
3. Volunteer
4. Delivery Partner
5. System Administrator
6. Food Safety Inspector

**Primary Use Cases:**

1. Register / Login
2. Add Food Donation
3. Inspect Food Donation
4. Accept / Reject Food Donation
5. Assign Pickup Task
6. Update Pickup Status
7. Accept Delivery Task
8. Deliver Food
9. Generate Reports
10. Track Donation Lifecycle

**3.5.1 UC-01 Register / Login**

| Field             | Details                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------|
| Use Case ID       | UC-01                                                                                                  |
| Use Case Name     | Register / Login                                                                                       |
| Created By        | Rudrakshi Chincholkar                                                                                  |
| Last Updated By   | Rudrakshi Chincholkar                                                                                  |
| Date Created      | Mar 2026                                                                                               |
| Date Last Updated | Mar 2026                                                                                               |
| Actor             | Food Donor, NGO / Food Bank, Volunteer, Delivery Partner, System Administrator, Food Safety Inspector  |
| Description       | Allows all authorized users to register a new account or log in to the system using valid credentials. |

|                                |                                                                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Preconditions</b>           | For login: user account must exist. For registration: valid user details must be provided.                                                                                                                    |
| <b>Postconditions</b>          | User is authenticated and redirected to their role-specific dashboard.                                                                                                                                        |
| <b>Priority</b>                | High                                                                                                                                                                                                          |
| <b>Frequency of Use</b>        | Very Frequent                                                                                                                                                                                                 |
| <b>Normal Course of Events</b> | 1. User opens the login or registration page. 2. User enters credentials or registration details. 3. System validates the input. 4. System retrieves the user role. 5. User is redirected to their dashboard. |
| <b>Alternative Courses</b>     | 1. User selects the Forgot Password option. 2. System initiates a password reset flow.                                                                                                                        |
| <b>Exceptions</b>              | 1. Incorrect username or password entered. 2. System displays an error message and denies access.                                                                                                             |
| <b>Includes</b>                | AuthenticationController                                                                                                                                                                                      |
| <b>Special Requirements</b>    | Passwords must be encrypted and stored securely using hashing.                                                                                                                                                |
| <b>Assumptions</b>             | User has valid credentials and an active account.                                                                                                                                                             |
| <b>Notes and Issues</b>        | All login attempts are logged for security auditing purposes.                                                                                                                                                 |

### 3.5.2 UC-02 Add Food Donation

| Field                  | Details               |
|------------------------|-----------------------|
| <b>Use Case ID</b>     | UC-02                 |
| <b>Use Case Name</b>   | Add Food request      |
| <b>Created By</b>      | Rudrakshi Chincholkar |
| <b>Last Updated By</b> | Rudrakshi Chincholkar |

|                                |                                                                                                                                                                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                                                                                                |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                                                                                                |
| <b>Actor</b>                   | Food Donor                                                                                                                                                                                                                                              |
| <b>Description</b>             | Allows a food donor to submit a new food donation with details such as food type, quantity, expiry date, and pickup location.                                                                                                                           |
| <b>Preconditions</b>           | Donor must be logged into the system.                                                                                                                                                                                                                   |
| <b>Postconditions</b>          | Food donation record is created and listed as available for NGO viewing.                                                                                                                                                                                |
| <b>Priority</b>                | High                                                                                                                                                                                                                                                    |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                                                                                                |
| <b>Normal Course of Events</b> | 1. Donor navigates to the Add Donation interface. 2. Donor enters food details including type, quantity, expiry, and location. 3. System validates the input against food safety guidelines. 4. System records the donation and notifies relevant NGOs. |
| <b>Alternative Courses</b>     | 1. Donor previews the donation before submitting. 2. System allows modification before final submission.                                                                                                                                                |
| <b>Exceptions</b>              | 1. Invalid or missing food details entered. 2. System displays a validation error and prevents submission.                                                                                                                                              |
| <b>Includes</b>                | Register / Login, Follow Food Safety Guidelines                                                                                                                                                                                                         |
| <b>Special Requirements</b>    | Donation records must include a timestamp and location for traceability.                                                                                                                                                                                |
| <b>Assumptions</b>             | Donor account is active and verified.                                                                                                                                                                                                                   |
| <b>Notes and Issues</b>        | This is the base use case for Edit Food Donation and Cancel Food Donation via «extend».                                                                                                                                                                 |

### 3.5.3 UC-03 Edit Food Donation

| Field                   | Details                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID             | UC-03                                                                                                                                                                                                |
| Use Case Name           | Edit Food request                                                                                                                                                                                    |
| Created By              | Rudrakshi Chincholkar                                                                                                                                                                                |
| Last Updated By         | Rudrakshi Chincholkar                                                                                                                                                                                |
| Date Created            | Mar 2026                                                                                                                                                                                             |
| Date Last Updated       | Mar 2026                                                                                                                                                                                             |
| Actor                   | Food Donor                                                                                                                                                                                           |
| Description             | Allows a food donor to modify the details of a previously submitted food donation if changes are needed. «extends» Add Food Donation.                                                                |
| Preconditions           | Donation must have been added and must not yet be accepted by an NGO.                                                                                                                                |
| Postconditions          | Donation record is updated with the corrected details.                                                                                                                                               |
| Priority                | Medium                                                                                                                                                                                               |
| Frequency of Use        | Occasional                                                                                                                                                                                           |
| Normal Course of Events | 1. Donor selects an existing donation from the dashboard. 2. Donor edits the required fields. 3. System validates the updated information. 4. System saves and displays the updated donation record. |
| Alternative Courses     | 1. Donor cancels the edit and retains the original details.                                                                                                                                          |
| Exceptions              | 1. Donation has already been accepted — editing is not permitted.<br>2. System displays an error message.                                                                                            |
| Includes                | Register / Login                                                                                                                                                                                     |
| Special Requirements    | Edit history should be logged for accountability.                                                                                                                                                    |

|                         |                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Assumptions</b>      | The donation is still in a pending or submitted state.                                                |
| <b>Notes and Issues</b> | This use case «extends» Add Food Donation and is only triggered when the donor needs to make changes. |

### 3.5.4 UC-04 Cancel Food Donation

| Field                          | Details                                                                                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>             | UC-04                                                                                                                                                                                     |
| <b>Use Case Name</b>           | Cancel Food request                                                                                                                                                                       |
| <b>Created By</b>              | Rudrakshi Chincholkar                                                                                                                                                                     |
| <b>Last Updated By</b>         | Rudrakshi Chincholkar                                                                                                                                                                     |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                                  |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                                  |
| <b>Actor</b>                   | Food Donor                                                                                                                                                                                |
| <b>Description</b>             | Allows a food donor to cancel a submitted food donation if they change their mind or if the food is no longer available. «extends» Add Food Donation.                                     |
| <b>Preconditions</b>           | Donation must exist and must not yet have been accepted or assigned.                                                                                                                      |
| <b>Postconditions</b>          | Donation is marked as cancelled and removed from the available listings.                                                                                                                  |
| <b>Priority</b>                | Medium                                                                                                                                                                                    |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                                                |
| <b>Normal Course of Events</b> | 1. Donor selects the donation from the dashboard. 2. Donor clicks Cancel Donation. 3. System prompts for a reason. 4. System marks the donation as cancelled and notifies any linked NGO. |

|                             |                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Alternative Courses</b>  | 1. Donor decides not to cancel and goes back to the dashboard.                                              |
| <b>Exceptions</b>           | 1. Donation has already been accepted — cancellation is not permitted. 2. System displays an error message. |
| <b>Includes</b>             | Register / Login                                                                                            |
| <b>Special Requirements</b> | Cancellation reason must be recorded for reporting purposes.                                                |
| <b>Assumptions</b>          | The donation is still in a submitted or pending state.                                                      |
| <b>Notes and Issues</b>     | This use case «extends» Add Food Donation and is only triggered when the donor changes their mind.          |

### 3.5.5 UC-05 View Food Request Status

| Field                    | Details                                                                                        |
|--------------------------|------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-05                                                                                          |
| <b>Use Case Name</b>     | View Food Request                                                                              |
| <b>Created By</b>        | Rudrakshi Chincholkar                                                                          |
| <b>Last Updated By</b>   | Rudrakshi Chincholkar                                                                          |
| <b>Date Created</b>      | Mar 2026                                                                                       |
| <b>Date Last Updated</b> | Mar 2026                                                                                       |
| <b>Actor</b>             | Food Donor                                                                                     |
| <b>Description</b>       | Allows a food donor to view the current lifecycle status of their submitted donation requests. |
| <b>Preconditions</b>     | Donor must be logged in and have at least one submitted donation.                              |
| <b>Postconditions</b>    | Current donation status is displayed to the donor.                                             |
| <b>Priority</b>          | Medium                                                                                         |

|                                |                                                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                      |
| <b>Normal Course of Events</b> | 1. Donor navigates to the My Donations page. 2. Donor selects a donation record. 3. System retrieves the current lifecycle stage. 4. System displays the status to the donor. |
| <b>Alternative Courses</b>     | 1. Donor filters donations by date or status type.                                                                                                                            |
| <b>Exceptions</b>              | 1. No donation records found. 2. System displays an appropriate message.                                                                                                      |
| <b>Includes</b>                | Register / Login, Track Donation Lifecycle                                                                                                                                    |
| <b>Special Requirements</b>    | Status must reflect real-time updates from volunteer and delivery partner actions.                                                                                            |
| <b>Assumptions</b>             | Donation records exist in the system.                                                                                                                                         |
| <b>Notes and Issues</b>        | This is the base use case for Feedback and Rating System via «extend».                                                                                                        |

### 3.5.6 UC-06 Feedback and Rating System

| Field                    | Details                    |
|--------------------------|----------------------------|
| <b>Use Case ID</b>       | UC-06                      |
| <b>Use Case Name</b>     | Feedback and Rating System |
| <b>Created By</b>        | Rudrakshi Chincholkar      |
| <b>Last Updated By</b>   | Rudrakshi Chincholkar      |
| <b>Date Created</b>      | Mar 2026                   |
| <b>Date Last Updated</b> | Mar 2026                   |
| <b>Actor</b>             | Food Donor                 |



|                                |                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>             | Allows a food donor to optionally submit feedback and a rating after viewing the completed status of their donation. «extends» View Food Request Status.                                             |
| <b>Preconditions</b>           | Donation must have reached a completed or delivered status.                                                                                                                                          |
| <b>Postconditions</b>          | Feedback and rating are recorded and linked to the donation record.                                                                                                                                  |
| <b>Priority</b>                | Low                                                                                                                                                                                                  |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                                                           |
| <b>Normal Course of Events</b> | 1. Donor views the completed donation status. 2. System prompts donor to submit optional feedback. 3. Donor enters a rating and comments. 4. System saves the feedback and links it to the donation. |
| <b>Alternative Courses</b>     | 1. Donor skips feedback — system does not require it.                                                                                                                                                |
| <b>Exceptions</b>              | 1. Donation is not yet completed. 2. Feedback option is not shown.                                                                                                                                   |
| <b>Includes</b>                | Register / Login                                                                                                                                                                                     |
| <b>Special Requirements</b>    | Feedback must be stored for NGO performance reporting.                                                                                                                                               |
| <b>Assumptions</b>             | Donation has been successfully delivered.                                                                                                                                                            |
| <b>Notes and Issues</b>        | This use case «extends» View Food Request Status and is only triggered optionally after completion.                                                                                                  |

### 3.5.7 UC-07 Follow Food Safety Guidelines

| Field                | Details                       |
|----------------------|-------------------------------|
| <b>Use Case ID</b>   | UC-07                         |
| <b>Use Case Name</b> | Follow Food Safety Guidelines |

|                                |                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Created By</b>              | Rudrakshi Chincholkar                                                                                                                              |
| <b>Last Updated By</b>         | Rudrakshi Chincholkar                                                                                                                              |
| <b>Date Created</b>            | Mar 2026                                                                                                                                           |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                           |
| <b>Actor</b>                   | Food Donor                                                                                                                                         |
| <b>Description</b>             | Allows a food donor to access and review food safety guidelines before submitting a donation to ensure compliance.                                 |
| <b>Preconditions</b>           | Donor must be logged in.                                                                                                                           |
| <b>Postconditions</b>          | Donor is informed of food safety requirements for donation.                                                                                        |
| <b>Priority</b>                | Medium                                                                                                                                             |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                           |
| <b>Normal Course of Events</b> | 1. Donor accesses the food safety guidelines section. 2. System displays the current safety rules. 3. Donor reads and acknowledges the guidelines. |
| <b>Alternative Courses</b>     | 1. Donor downloads the guidelines as a PDF.                                                                                                        |
| <b>Exceptions</b>              | 1. Guidelines not available — system displays an error.                                                                                            |
| <b>Includes</b>                | Register / Login                                                                                                                                   |
| <b>Special Requirements</b>    | Guidelines must be regularly updated by the Food Safety Inspector.                                                                                 |
| <b>Assumptions</b>             | Food safety rules are defined and maintained in the system.                                                                                        |
| <b>Notes and Issues</b>        | This is a standalone use case linked to Add Food Donation to promote compliance.                                                                   |

### 3.5.8 UC-08 View Available Donations

| Field                   | Details                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID             | UC-08                                                                                                                                                                                |
| Use Case Name           | View Available Donations                                                                                                                                                             |
| Created By              | Rudrakshi Chincholkar                                                                                                                                                                |
| Last Updated By         | Rudrakshi Chincholkar                                                                                                                                                                |
| Date Created            | Mar 2026                                                                                                                                                                             |
| Date Last Updated       | Mar 2026                                                                                                                                                                             |
| Actor                   | NGO / Food Bank                                                                                                                                                                      |
| Description             | Allows an NGO to browse all currently available and unaccepted food donations listed by donors.                                                                                      |
| Preconditions           | NGO must be logged in. At least one donation must be available.                                                                                                                      |
| Postconditions          | List of available donations is displayed to the NGO.                                                                                                                                 |
| Priority                | High                                                                                                                                                                                 |
| Frequency of Use        | Frequent                                                                                                                                                                             |
| Normal Course of Events | 1. NGO navigates to the Available Donations page. 2. System retrieves all unaccepted donations. 3. System displays donations with details such as food type, quantity, and location. |
| Alternative Courses     | 1. NGO filters donations by type, location, or date.                                                                                                                                 |
| Exceptions              | 1. No donations available. 2. System displays an appropriate message.                                                                                                                |
| Includes                | Register / Login                                                                                                                                                                     |
| Special Requirements    | Real-time availability must be maintained.                                                                                                                                           |

|                         |                                                                       |
|-------------------------|-----------------------------------------------------------------------|
| <b>Assumptions</b>      | Donors have submitted valid listings.                                 |
| <b>Notes and Issues</b> | This is the base use case for Donation Matching System via «include». |

### 3.5.9 UC-09 Accept Food Donation

| Field                          | Details                                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>             | UC-09                                                                                                                                                                    |
| <b>Use Case Name</b>           | Accept Food Donation                                                                                                                                                     |
| <b>Created By</b>              | Rudrakshi Chincholkar                                                                                                                                                    |
| <b>Last Updated By</b>         | Rudrakshi Chincholkar                                                                                                                                                    |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                 |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                 |
| <b>Actor</b>                   | NGO / Food Bank                                                                                                                                                          |
| <b>Description</b>             | Allows an NGO to formally accept a food donation after a successful inspection, initiating the pickup workflow.                                                          |
| <b>Preconditions</b>           | Donation must have been inspected and cleared by a food safety inspector.                                                                                                |
| <b>Postconditions</b>          | Donation status is updated to Accepted and the pickup assignment process begins.                                                                                         |
| <b>Priority</b>                | High                                                                                                                                                                     |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                 |
| <b>Normal Course of Events</b> | 1. NGO reviews the inspected donation. 2. NGO clicks Accept Donation. 3. System updates the donation status to Accepted. 4. System triggers volunteer pickup assignment. |

|                             |                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------|
| <b>Alternative Courses</b>  | 1. NGO requests additional inspection before deciding.                                 |
| <b>Exceptions</b>           | 1. Donation has not been inspected. 2. System blocks acceptance and displays an error. |
| <b>Includes</b>             | Register / Login, Inspect Food Donation                                                |
| <b>Special Requirements</b> | Acceptance must be logged with a timestamp for lifecycle tracking.                     |
| <b>Assumptions</b>          | Inspection has been completed and the food is safe.                                    |
| <b>Notes and Issues</b>     | Acceptance «includes» Inspect Food Donation — inspection is mandatory.                 |

### 3.5.10 UC-10 Reject Food Donation

| Field                    | Details                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-10                                                                                                     |
| <b>Use Case Name</b>     | Reject Food Donation                                                                                      |
| <b>Created By</b>        | Rudrakshi Chincholkar                                                                                     |
| <b>Last Updated By</b>   | Rudrakshi Chincholkar                                                                                     |
| <b>Date Created</b>      | Mar 2026                                                                                                  |
| <b>Date Last Updated</b> | Mar 2026                                                                                                  |
| <b>Actor</b>             | NGO / Food Bank                                                                                           |
| <b>Description</b>       | Allows an NGO to reject a food donation after inspection if it does not meet safety or quality standards. |
| <b>Preconditions</b>     | Donation must have been inspected.                                                                        |
| <b>Postconditions</b>    | Donation is marked as Rejected and the donor is notified.                                                 |

|                                |                                                                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Priority</b>                | High                                                                                                                                                                 |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                           |
| <b>Normal Course of Events</b> | 1. NGO reviews the inspection result. 2. NGO selects Reject Donation. 3. NGO enters a rejection reason. 4. System updates status to Rejected and notifies the donor. |
| <b>Alternative Courses</b>     | 1. NGO escalates the rejection for penalization of the donor.                                                                                                        |
| <b>Exceptions</b>              | 1. Rejection reason not provided. 2. System requires a reason before proceeding.                                                                                     |
| <b>Includes</b>                | Register / Login, Inspect Food Donation                                                                                                                              |
| <b>Special Requirements</b>    | Rejection reason must be recorded for audit purposes.                                                                                                                |
| <b>Assumptions</b>             | Inspection result is available and indicates non-compliance.                                                                                                         |
| <b>Notes and Issues</b>        | Rejection «includes» Inspect Food Donation — inspection result drives the rejection decision.                                                                        |

### 3.5.11 UC-11 Manage Volunteers

| Field                    | Details           |
|--------------------------|-------------------|
| <b>Use Case ID</b>       | UC-11             |
| <b>Use Case Name</b>     | Manage Volunteers |
| <b>Created By</b>        | Nihira Neralwar   |
| <b>Last Updated By</b>   | Nihira Neralwar   |
| <b>Date Created</b>      | Mar 2026          |
| <b>Date Last Updated</b> | Mar 2026          |
| <b>Actor</b>             | NGO / Food Bank   |

|                                |                                                                                                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>             | Allows an NGO to view, add, remove, and manage volunteers registered under their organization.                                                                                                |
| <b>Preconditions</b>           | NGO must be logged in.                                                                                                                                                                        |
| <b>Postconditions</b>          | Volunteer records are updated in the system.                                                                                                                                                  |
| <b>Priority</b>                | High                                                                                                                                                                                          |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                                      |
| <b>Normal Course of Events</b> | 1. NGO navigates to the Volunteer Management panel. 2. NGO views the current list of volunteers. 3. NGO adds or removes volunteers as required. 4. System saves the updated volunteer roster. |
| <b>Alternative Courses</b>     | 1. NGO updates volunteer availability status.                                                                                                                                                 |
| <b>Exceptions</b>              | 1. Invalid volunteer details provided. 2. System displays a validation error.                                                                                                                 |
| <b>Includes</b>                | Register / Login, Assign Pickup Task                                                                                                                                                          |
| <b>Special Requirements</b>    | Volunteer records must include availability and contact information.                                                                                                                          |
| <b>Assumptions</b>             | Volunteers have registered accounts in the system.                                                                                                                                            |
| <b>Notes and Issues</b>        | Managing volunteers «includes» Assign Pickup Task as part of the coordination workflow.                                                                                                       |

### 3.5.12 UC-12 Manage Delivery Partners

| Field                  | Details                  |
|------------------------|--------------------------|
| <b>Use Case ID</b>     | UC-12                    |
| <b>Use Case Name</b>   | Manage Delivery Partners |
| <b>Created By</b>      | Nihira Neralwar          |
| <b>Last Updated By</b> | Nihira Neralwar          |

|                                |                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                                 |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                                 |
| <b>Actor</b>                   | NGO / Food Bank                                                                                                                                                                          |
| <b>Description</b>             | Allows an NGO to manage the delivery partners associated with their food distribution operations.                                                                                        |
| <b>Preconditions</b>           | NGO must be logged in.                                                                                                                                                                   |
| <b>Postconditions</b>          | Delivery partner records are updated in the system.                                                                                                                                      |
| <b>Priority</b>                | Medium                                                                                                                                                                                   |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                                               |
| <b>Normal Course of Events</b> | 1. NGO opens the Delivery Partner Management section. 2. NGO views the current list of delivery partners. 3. NGO adds, removes, or updates partner information. 4. System saves changes. |
| <b>Alternative Courses</b>     | 1. NGO assigns a delivery partner directly to a donation.                                                                                                                                |
| <b>Exceptions</b>              | 1. Partner details incomplete. 2. System displays an error and requests correct information.                                                                                             |
| <b>Includes</b>                | Register / Login                                                                                                                                                                         |
| <b>Special Requirements</b>    | Delivery partner records must include vehicle and contact information.                                                                                                                   |
| <b>Assumptions</b>             | Delivery partners have registered accounts in the system.                                                                                                                                |
| <b>Notes and Issues</b>        | Standalone use case for managing the delivery partner pool.                                                                                                                              |

### 3.5.13 UC-13 Penalize Food Donor

| Field | Details |
|-------|---------|
|-------|---------|



|                                |                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>             | UC-13                                                                                                                                                                                |
| <b>Use Case Name</b>           | Penalize Food Donor                                                                                                                                                                  |
| <b>Created By</b>              | Nihira Neralwar                                                                                                                                                                      |
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                                                                      |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                             |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                             |
| <b>Actor</b>                   | NGO / Food Bank                                                                                                                                                                      |
| <b>Description</b>             | Allows an NGO to issue a penalty to a food donor when a severe food safety violation is detected during inspection. «extends» Inspect Food Donation.                                 |
| <b>Preconditions</b>           | Inspection must have found a severe safety violation.                                                                                                                                |
| <b>Postconditions</b>          | Donor is penalized and a record is maintained.                                                                                                                                       |
| <b>Priority</b>                | Medium                                                                                                                                                                               |
| <b>Frequency of Use</b>        | Rare                                                                                                                                                                                 |
| <b>Normal Course of Events</b> | 1. Inspection reveals a severe violation. 2. NGO selects Penalize Donor. 3. NGO enters the violation details and penalty type. 4. System records the penalty and notifies the donor. |
| <b>Alternative Courses</b>     | 1. NGO issues a warning instead of a formal penalty.                                                                                                                                 |
| <b>Exceptions</b>              | 1. No inspection record found. 2. System blocks the penalization.                                                                                                                    |
| <b>Includes</b>                | Register / Login                                                                                                                                                                     |
| <b>Special Requirements</b>    | Penalty records must be logged and linked to the inspection report.                                                                                                                  |
| <b>Assumptions</b>             | Inspection was conducted and violation was documented.                                                                                                                               |

|                         |                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Notes and Issues</b> | This use case «extends» Inspect Food Donation and is triggered only when a severe violation is found. |
|-------------------------|-------------------------------------------------------------------------------------------------------|

### 3.5.14 UC-14 Assign / Manage Food Inspector

| Field                          | Details                                                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>             | UC-14                                                                                                                                                              |
| <b>Use Case Name</b>           | Assign / Manage Food Inspector                                                                                                                                     |
| <b>Created By</b>              | Nihira Neralwar                                                                                                                                                    |
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                                                    |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                           |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                           |
| <b>Actor</b>                   | NGO / Food Bank                                                                                                                                                    |
| <b>Description</b>             | Allows an NGO to assign food safety inspectors to specific donations and manage their inspector roster.                                                            |
| <b>Preconditions</b>           | NGO must be logged in. Food safety inspectors must have registered accounts.                                                                                       |
| <b>Postconditions</b>          | Inspector is assigned to the donation and notified.                                                                                                                |
| <b>Priority</b>                | High                                                                                                                                                               |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                           |
| <b>Normal Course of Events</b> | 1. NGO selects a pending donation. 2. NGO assigns an available food safety inspector. 3. System notifies the inspector. 4. Inspector proceeds with the inspection. |
| <b>Alternative Courses</b>     | 1. NGO reassigns a different inspector if the first is unavailable.                                                                                                |
| <b>Exceptions</b>              | 1. No inspectors are available. 2. System alerts the NGO to the unavailability.                                                                                    |

|                             |                                                              |
|-----------------------------|--------------------------------------------------------------|
| <b>Includes</b>             | Register / Login                                             |
| <b>Special Requirements</b> | Inspector assignment must be logged with a timestamp.        |
| <b>Assumptions</b>          | At least one food safety inspector is registered and active. |
| <b>Notes and Issues</b>     | Standalone management use case for inspector assignment.     |

### 3.5.15 UC-15 Manage Unsafe Food

| Field                    | Details                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-15                                                                                                                                        |
| <b>Use Case Name</b>     | Manage Unsafe Food                                                                                                                           |
| <b>Created By</b>        | Nihira Neralwar                                                                                                                              |
| <b>Last Updated By</b>   | Nihira Neralwar                                                                                                                              |
| <b>Date Created</b>      | Mar 2026                                                                                                                                     |
| <b>Date Last Updated</b> | Mar 2026                                                                                                                                     |
| <b>Actor</b>             | NGO / Food Bank                                                                                                                              |
| <b>Description</b>       | Allows an NGO to handle food donations that have been flagged as unsafe, including arranging safe disposal. «extends» Inspect Food Donation. |
| <b>Preconditions</b>     | Food donation must have been flagged as unsafe by a food safety inspector.                                                                   |
| <b>Postconditions</b>    | Unsafe food is safely disposed of and the record is updated.                                                                                 |
| <b>Priority</b>          | High                                                                                                                                         |
| <b>Frequency of Use</b>  | Occasional                                                                                                                                   |

|                                |                                                                                                                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Normal Course of Events</b> | 1. NGO receives an unsafe food flag notification. 2. NGO opens the unsafe food management module. 3. NGO arranges for safe disposal of the donation. 4. System updates the donation record to Disposed. |
| <b>Alternative Courses</b>     | 1. NGO escalates to health authorities if needed.                                                                                                                                                       |
| <b>Exceptions</b>              | 1. Disposal arrangement fails. 2. System logs the error and alerts admin.                                                                                                                               |
| <b>Includes</b>                | Register / Login                                                                                                                                                                                        |
| <b>Special Requirements</b>    | Disposal must be logged with method, date, and responsible officer.                                                                                                                                     |
| <b>Assumptions</b>             | Food has been flagged by the inspector before this use case is triggered.                                                                                                                               |
| <b>Notes and Issues</b>        | This use case «extends» Inspect Food Donation and is triggered only when food is found contaminated.                                                                                                    |

### 3.5.16 UC-16 Donation Matching System

| Field                    | Details                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-16                                                                                                                                               |
| <b>Use Case Name</b>     | Donation Matching System                                                                                                                            |
| <b>Created By</b>        | Nihira Neralwar                                                                                                                                     |
| <b>Last Updated By</b>   | Nihira Neralwar                                                                                                                                     |
| <b>Date Created</b>      | Mar 2026                                                                                                                                            |
| <b>Date Last Updated</b> | Mar 2026                                                                                                                                            |
| <b>Actor</b>             | NGO / Food Bank                                                                                                                                     |
| <b>Description</b>       | Automatically matches available food donations to NGO requirements based on food type, quantity, and location. «includes» View Available Donations. |

|                                |                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Preconditions</b>           | Donation listings must be available and NGO requirements must be defined.                                                                                                              |
| <b>Postconditions</b>          | A suggested donation match is presented to the NGO.                                                                                                                                    |
| <b>Priority</b>                | High                                                                                                                                                                                   |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                               |
| <b>Normal Course of Events</b> | 1. NGO initiates the donation matching process. 2. System scans all available donations. 3. System applies matching criteria. 4. System presents the best-matched donation to the NGO. |
| <b>Alternative Courses</b>     | 1. NGO manually overrides the match and selects a different donation.                                                                                                                  |
| <b>Exceptions</b>              | 1. No suitable match found. 2. System notifies the NGO and suggests broadening criteria.                                                                                               |
| <b>Includes</b>                | Register / Login, View Available Donations                                                                                                                                             |
| <b>Special Requirements</b>    | Matching algorithm must consider food type, quantity, expiry, and proximity.                                                                                                           |
| <b>Assumptions</b>             | Sufficient donation listings exist for matching.                                                                                                                                       |
| <b>Notes and Issues</b>        | Matching «includes» View Available Donations as the matching process requires scanning the full availability list.                                                                     |

### 3.5.17 UC-17 View Pickup Requests

| Field                  | Details              |
|------------------------|----------------------|
| <b>Use Case ID</b>     | UC-17                |
| <b>Use Case Name</b>   | View Pickup Requests |
| <b>Created By</b>      | Nihira Neralwar      |
| <b>Last Updated By</b> | Nihira Neralwar      |

|                                |                                                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                       |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                       |
| <b>Actor</b>                   | Volunteer                                                                                                                                                                      |
| <b>Description</b>             | Allows a volunteer to view all currently open pickup requests assigned to or available for them.                                                                               |
| <b>Preconditions</b>           | Volunteer must be logged in and have set their availability.                                                                                                                   |
| <b>Postconditions</b>          | List of available pickup requests is displayed.                                                                                                                                |
| <b>Priority</b>                | High                                                                                                                                                                           |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                       |
| <b>Normal Course of Events</b> | 1. Volunteer navigates to the Pickup Requests page. 2. System retrieves all open and relevant pickup requests. 3. System displays the list with donation details and location. |
| <b>Alternative Courses</b>     | 1. Volunteer filters requests by location or food type.                                                                                                                        |
| <b>Exceptions</b>              | 1. No pickup requests available. 2. System displays an appropriate message.                                                                                                    |
| <b>Includes</b>                | Register / Login                                                                                                                                                               |
| <b>Special Requirements</b>    | Requests must show real-time distance and donation details.                                                                                                                    |
| <b>Assumptions</b>             | NGO has assigned pickup tasks linked to accepted donations.                                                                                                                    |
| <b>Notes and Issues</b>        | This is the base use case for Accept Pickup Task via «include».                                                                                                                |

### 3.5.18 UC-18 Accept Pickup Task

| Field | Details |
|-------|---------|
|-------|---------|

|                                |                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>             | UC-18                                                                                                                                                                   |
| <b>Use Case Name</b>           | Accept Pickup Task                                                                                                                                                      |
| <b>Created By</b>              | Nihira Neralwar                                                                                                                                                         |
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                                                         |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                |
| <b>Actor</b>                   | Volunteer                                                                                                                                                               |
| <b>Description</b>             | Allows a volunteer to accept a specific pickup task from the available request list. «includes» View Pickup Requests.                                                   |
| <b>Preconditions</b>           | Volunteer must be logged in and have viewed pickup requests.                                                                                                            |
| <b>Postconditions</b>          | Pickup task is assigned to the volunteer and the donation status is updated.                                                                                            |
| <b>Priority</b>                | High                                                                                                                                                                    |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                |
| <b>Normal Course of Events</b> | 1. Volunteer views the pickup request list. 2. Volunteer selects a request. 3. Volunteer clicks Accept Task. 4. System assigns the task and notifies the donor and NGO. |
| <b>Alternative Courses</b>     | 1. Volunteer declines a request and returns to the list.                                                                                                                |
| <b>Exceptions</b>              | 1. Task already accepted by another volunteer. 2. System notifies the volunteer and refreshes the list.                                                                 |
| <b>Includes</b>                | Register / Login, View Pickup Requests                                                                                                                                  |
| <b>Special Requirements</b>    | Task acceptance must be confirmed within a time window.                                                                                                                 |
| <b>Assumptions</b>             | Pickup requests are available and unassigned.                                                                                                                           |

|                         |                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------|
| <b>Notes and Issues</b> | This use case «includes» View Pickup Requests as a volunteer must see the requests to accept one. |
|-------------------------|---------------------------------------------------------------------------------------------------|

### 3.5.19 UC-19 Set Availability

| Field                          | Details                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>             | UC-19                                                                                                                                                     |
| <b>Use Case Name</b>           | Set Availability                                                                                                                                          |
| <b>Created By</b>              | Nihira Neralwar                                                                                                                                           |
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                                           |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                  |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                  |
| <b>Actor</b>                   | Volunteer                                                                                                                                                 |
| <b>Description</b>             | Allows a volunteer to set or update their availability schedule so the system can assign appropriate pickup tasks.                                        |
| <b>Preconditions</b>           | Volunteer must be logged in.                                                                                                                              |
| <b>Postconditions</b>          | Volunteer availability is updated in the system.                                                                                                          |
| <b>Priority</b>                | Medium                                                                                                                                                    |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                |
| <b>Normal Course of Events</b> | 1. Volunteer navigates to the Availability Settings page. 2. Volunteer selects available dates and time slots. 3. System saves the availability schedule. |
| <b>Alternative Courses</b>     | 1. Volunteer marks themselves as temporarily unavailable.                                                                                                 |
| <b>Exceptions</b>              | 1. No time slots selected. 2. System prompts volunteer to select at least one slot.                                                                       |



|                             |                                                                |
|-----------------------------|----------------------------------------------------------------|
| <b>Includes</b>             | Register / Login                                               |
| <b>Special Requirements</b> | Availability must be visible to the NGO for task assignment.   |
| <b>Assumptions</b>          | Volunteer is active and willing to take tasks.                 |
| <b>Notes and Issues</b>     | Standalone use case that enables the task assignment workflow. |

### 3.5.20 UC-20 Update Pickup Status

| Field                    | Details                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-20                                                                                                  |
| <b>Use Case Name</b>     | Update Pickup Status                                                                                   |
| <b>Created By</b>        | Nihira Neralwar                                                                                        |
| <b>Last Updated By</b>   | Nihira Neralwar                                                                                        |
| <b>Date Created</b>      | Mar 2026                                                                                               |
| <b>Date Last Updated</b> | Mar 2026                                                                                               |
| <b>Actor</b>             | Volunteer                                                                                              |
| <b>Description</b>       | Allows a volunteer to update the status of an ongoing pickup task as it progresses through the stages. |
| <b>Preconditions</b>     | Volunteer must have accepted a pickup task.                                                            |
| <b>Postconditions</b>    | Pickup status is updated and all relevant parties are notified.                                        |
| <b>Priority</b>          | High                                                                                                   |
| <b>Frequency of Use</b>  | Frequent                                                                                               |

|                                |                                                                                                                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Normal Course of Events</b> | 1. Volunteer opens the active pickup task. 2. Volunteer selects the new status such as On the Way or Picked Up. 3. System updates the donation lifecycle status. 4. System notifies the NGO and donor. |
| <b>Alternative Courses</b>     | 1. Volunteer adds notes or photos as part of the update.                                                                                                                                               |
| <b>Exceptions</b>              | 1. Network failure prevents the status update. 2. System queues the update and retries.                                                                                                                |
| <b>Includes</b>                | Register / Login                                                                                                                                                                                       |
| <b>Special Requirements</b>    | Status updates must be timestamped and linked to the donation record.                                                                                                                                  |
| <b>Assumptions</b>             | Volunteer has an active accepted pickup task.                                                                                                                                                          |
| <b>Notes and Issues</b>        | This is the base use case for Report Issues in Pickup via «extend».                                                                                                                                    |

### 3.5.21 UC-21 Report Issues in Pickup

| Field                    | Details                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-21                                                                                                         |
| <b>Use Case Name</b>     | Report Issues in Pickup                                                                                       |
| <b>Created By</b>        | Nihira Neralwar                                                                                               |
| <b>Last Updated By</b>   | Nihira Neralwar                                                                                               |
| <b>Date Created</b>      | Mar 2026                                                                                                      |
| <b>Date Last Updated</b> | Mar 2026                                                                                                      |
| <b>Actor</b>             | Volunteer                                                                                                     |
| <b>Description</b>       | Allows a volunteer to report a problem encountered during the pickup process. «extends» Update Pickup Status. |

|                                |                                                                                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Preconditions</b>           | Volunteer must be performing an active pickup task.                                                                                                                                        |
| <b>Postconditions</b>          | Issue is logged and the NGO is notified for corrective action.                                                                                                                             |
| <b>Priority</b>                | Medium                                                                                                                                                                                     |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                                                 |
| <b>Normal Course of Events</b> | 1. Volunteer encounters an issue during pickup. 2. Volunteer opens the Report Issue option. 3. Volunteer enters a description of the problem. 4. System logs the issue and alerts the NGO. |
| <b>Alternative Courses</b>     | 1. Volunteer attaches a photo as evidence of the issue.                                                                                                                                    |
| <b>Exceptions</b>              | 1. Issue description is blank. 2. System requires a description before submission.                                                                                                         |
| <b>Includes</b>                | Register / Login                                                                                                                                                                           |
| <b>Special Requirements</b>    | Issues must be logged with timestamp, location, and description.                                                                                                                           |
| <b>Assumptions</b>             | Volunteer has an active pickup task in progress.                                                                                                                                           |
| <b>Notes and Issues</b>        | This use case «extends» Update Pickup Status and is triggered only when an issue arises.                                                                                                   |

### 3.5.22 UC-22 Accept Delivery Task

| Field                  | Details              |
|------------------------|----------------------|
| <b>Use Case ID</b>     | UC-22                |
| <b>Use Case Name</b>   | Accept Delivery Task |
| <b>Created By</b>      | Nihira Neralwar      |
| <b>Last Updated By</b> | Nihira Neralwar      |
| <b>Date Created</b>    | Mar 2026             |

|                                |                                                                                                                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                                                        |
| <b>Actor</b>                   | Delivery Partner                                                                                                                                                                                                |
| <b>Description</b>             | Allows a delivery partner to accept an assigned delivery task for a food donation.                                                                                                                              |
| <b>Preconditions</b>           | Delivery partner must be logged in and a delivery task must be assigned.                                                                                                                                        |
| <b>Postconditions</b>          | Delivery task is confirmed and the pickup can begin.                                                                                                                                                            |
| <b>Priority</b>                | High                                                                                                                                                                                                            |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                                                        |
| <b>Normal Course of Events</b> | 1. Delivery partner views the assigned delivery task. 2. Partner reviews the details including pickup location and destination. 3. Partner accepts the task. 4. System updates the delivery status to Accepted. |
| <b>Alternative Courses</b>     | 1. Partner declines the task — system reassigns to another partner.                                                                                                                                             |
| <b>Exceptions</b>              | 1. Task has been cancelled. 2. System notifies the partner and removes the task.                                                                                                                                |
| <b>Includes</b>                | Register / Login                                                                                                                                                                                                |
| <b>Special Requirements</b>    | Task acceptance must be recorded with a timestamp.                                                                                                                                                              |
| <b>Assumptions</b>             | A delivery task has been assigned by the NGO.                                                                                                                                                                   |
| <b>Notes and Issues</b>        | This is the base use case for Pick Up Food via «include».                                                                                                                                                       |

### 3.5.23 UC-23 Pick Up Food

| Field              | Details |
|--------------------|---------|
| <b>Use Case ID</b> | UC-23   |

|                                |                                                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name</b>           | Pick Up Food                                                                                                                                                                   |
| <b>Created By</b>              | Nihira Neralwar                                                                                                                                                                |
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                                                                |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                       |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                       |
| <b>Actor</b>                   | Delivery Partner                                                                                                                                                               |
| <b>Description</b>             | Allows a delivery partner to confirm the physical pickup of food from the volunteer handoff point. «includes» Accept Delivery Task.                                            |
| <b>Preconditions</b>           | Delivery task must have been accepted by the delivery partner.                                                                                                                 |
| <b>Postconditions</b>          | Food pickup is confirmed and delivery can begin.                                                                                                                               |
| <b>Priority</b>                | High                                                                                                                                                                           |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                       |
| <b>Normal Course of Events</b> | 1. Delivery partner arrives at the pickup location. 2. Partner confirms food collection in the system. 3. System updates status to Food Picked Up. 4. System notifies the NGO. |
| <b>Alternative Courses</b>     | 1. Partner reports a discrepancy in the food quantity.                                                                                                                         |
| <b>Exceptions</b>              | 1. Food not available at pickup location. 2. System logs the issue and alerts the NGO.                                                                                         |
| <b>Includes</b>                | Register / Login, Accept Delivery Task                                                                                                                                         |
| <b>Special Requirements</b>    | Pickup confirmation must include location verification.                                                                                                                        |
| <b>Assumptions</b>             | Food is ready and available at the specified pickup location.                                                                                                                  |

|                         |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| <b>Notes and Issues</b> | This use case «includes» Accept Delivery Task as pickup cannot happen without prior task acceptance. |
|-------------------------|------------------------------------------------------------------------------------------------------|

### 3.5.24 UC-24 Deliver Food

| Field                          | Details                                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>             | UC-24                                                                                                                                                                                                                   |
| <b>Use Case Name</b>           | Deliver Food                                                                                                                                                                                                            |
| <b>Created By</b>              | Nihira Neralwar                                                                                                                                                                                                         |
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                                                                                                         |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                                                                                |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                                                                                |
| <b>Actor</b>                   | Delivery Partner                                                                                                                                                                                                        |
| <b>Description</b>             | Allows a delivery partner to confirm successful delivery of food to the designated beneficiary location. «includes» Pick Up Food.                                                                                       |
| <b>Preconditions</b>           | Food must have been picked up and confirmed in the system.                                                                                                                                                              |
| <b>Postconditions</b>          | Delivery is confirmed and donation lifecycle advances to Delivered.                                                                                                                                                     |
| <b>Priority</b>                | High                                                                                                                                                                                                                    |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                                                                |
| <b>Normal Course of Events</b> | 1. Delivery partner arrives at the destination. 2. Partner delivers the food to the beneficiary. 3. Partner confirms delivery in the system. 4. System updates donation status to Delivered and notifies NGO and donor. |

|                             |                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Alternative Courses</b>  | 1. Beneficiary is not available — partner reschedules delivery.                                     |
| <b>Exceptions</b>           | 1. Delivery location not found. 2. System allows partner to report the issue.                       |
| <b>Includes</b>             | Register / Login, Pick Up Food                                                                      |
| <b>Special Requirements</b> | Delivery confirmation must include timestamp and destination verification.                          |
| <b>Assumptions</b>          | Food has been successfully picked up and is in transit.                                             |
| <b>Notes and Issues</b>     | This use case «includes» Pick Up Food as delivery cannot happen without the pickup being completed. |

### 3.5.25 UC-25 Update Delivery Status

| Field                    | Details                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-25                                                                                                          |
| <b>Use Case Name</b>     | Update Delivery Status                                                                                         |
| <b>Created By</b>        | Nihira Neralwar                                                                                                |
| <b>Last Updated By</b>   | Nihira Neralwar                                                                                                |
| <b>Date Created</b>      | Mar 2026                                                                                                       |
| <b>Date Last Updated</b> | Mar 2026                                                                                                       |
| <b>Actor</b>             | Delivery Partner                                                                                               |
| <b>Description</b>       | Allows a delivery partner to update the current status of a delivery as it progresses. «extends» Deliver Food. |
| <b>Preconditions</b>     | Delivery partner must have an active delivery task in progress.                                                |
| <b>Postconditions</b>    | Delivery status is updated and stakeholders are notified.                                                      |

|                                |                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Priority</b>                | Medium                                                                                                                                                                  |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                                |
| <b>Normal Course of Events</b> | 1. Delivery partner opens the active delivery task. 2. Partner selects a new status such as In Transit or Delivered. 3. System records the update and notifies the NGO. |
| <b>Alternative Courses</b>     | 1. Partner adds notes or photos to the status update.                                                                                                                   |
| <b>Exceptions</b>              | 1. Connectivity issues prevent the update. 2. System queues and retries.                                                                                                |
| <b>Includes</b>                | Register / Login                                                                                                                                                        |
| <b>Special Requirements</b>    | All status updates must be timestamped and linked to the delivery record.                                                                                               |
| <b>Assumptions</b>             | An active delivery task exists.                                                                                                                                         |
| <b>Notes and Issues</b>        | This use case «extends» Deliver Food and is triggered as the delivery progresses through stages.                                                                        |

### 3.5.26 UC-26 Manage Users

| Field                    | Details              |
|--------------------------|----------------------|
| <b>Use Case ID</b>       | UC-26                |
| <b>Use Case Name</b>     | Manage Users         |
| <b>Created By</b>        | Nihira Neralwar      |
| <b>Last Updated By</b>   | Nihira Neralwar      |
| <b>Date Created</b>      | Mar 2026             |
| <b>Date Last Updated</b> | Mar 2026             |
| <b>Actor</b>             | System Administrator |



|                                |                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>             | Allows the system administrator to create, update, deactivate, or remove user accounts across all roles.                                                                               |
| <b>Preconditions</b>           | Administrator must be logged in.                                                                                                                                                       |
| <b>Postconditions</b>          | User accounts are created, updated, or deactivated as required.                                                                                                                        |
| <b>Priority</b>                | High                                                                                                                                                                                   |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                                             |
| <b>Normal Course of Events</b> | 1. Admin accesses the User Management panel. 2. Admin selects Create, Edit, or Deactivate. 3. Admin enters or updates user details. 4. System saves the changes and notifies the user. |
| <b>Alternative Courses</b>     | 1. Admin searches for a user by name or role.                                                                                                                                          |
| <b>Exceptions</b>              | 1. Invalid user data entered. 2. System displays a validation error.                                                                                                                   |
| <b>Includes</b>                | Register / Login                                                                                                                                                                       |
| <b>Special Requirements</b>    | User access must comply with role-based security policies.                                                                                                                             |
| <b>Assumptions</b>             | Administrator has full system privileges.                                                                                                                                              |
| <b>Notes and Issues</b>        | User roles determine system access permissions across all modules.                                                                                                                     |

### 3.5.27 UC-27 Monitor System Activity

| Field                | Details                 |
|----------------------|-------------------------|
| <b>Use Case ID</b>   | UC-27                   |
| <b>Use Case Name</b> | Monitor System Activity |
| <b>Created By</b>    | Nihira Neralwar         |

|                                |                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                             |
| <b>Date Created</b>            | Mar 2026                                                                                                                                    |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                    |
| <b>Actor</b>                   | System Administrator                                                                                                                        |
| <b>Description</b>             | Allows the system administrator to monitor real-time system activities including user actions, donation events, and system errors.          |
| <b>Preconditions</b>           | Administrator must be logged in.                                                                                                            |
| <b>Postconditions</b>          | Activity dashboard is displayed with current system events.                                                                                 |
| <b>Priority</b>                | High                                                                                                                                        |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                    |
| <b>Normal Course of Events</b> | 1. Admin navigates to the Activity Monitoring panel. 2. System displays real-time events and logs. 3. Admin reviews activity for anomalies. |
| <b>Alternative Courses</b>     | 1. Admin filters activity by user, role, or event type.                                                                                     |
| <b>Exceptions</b>              | 1. Logging system is unavailable. 2. System alerts the admin.                                                                               |
| <b>Includes</b>                | Register / Login                                                                                                                            |
| <b>Special Requirements</b>    | Activity logs must be tamper-resistant and timestamped.                                                                                     |
| <b>Assumptions</b>             | Logging is continuously active throughout system operation.                                                                                 |
| <b>Notes and Issues</b>        | Standalone use case for ongoing system oversight.                                                                                           |

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### 3.5.28 UC-28 Generate Reports

| Field                   | Details                                                                                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID             | UC-28                                                                                                                                                                                    |
| Use Case Name           | Generate Reports                                                                                                                                                                         |
| Created By              | Nihira Neralwar                                                                                                                                                                          |
| Last Updated By         | Nihira Neralwar                                                                                                                                                                          |
| Date Created            | Mar 2026                                                                                                                                                                                 |
| Date Last Updated       | Mar 2026                                                                                                                                                                                 |
| Actor                   | System Administrator                                                                                                                                                                     |
| Description             | Allows the system administrator to generate operational reports on donation activity, user performance, and lifecycle metrics.<br>«includes» Track Donation Lifecycle.                   |
| Preconditions           | Administrator must be logged in and donation data must exist.                                                                                                                            |
| Postconditions          | Report is generated and made available for download or viewing.                                                                                                                          |
| Priority                | Medium                                                                                                                                                                                   |
| Frequency of Use        | Periodic                                                                                                                                                                                 |
| Normal Course of Events | 1. Admin selects the Report Generation module. 2. Admin chooses report type and date range. 3. System retrieves lifecycle and activity data. 4. System compiles and displays the report. |
| Alternative Courses     | 1. Admin exports the report as a PDF or CSV file.                                                                                                                                        |
| Exceptions              | 1. Insufficient data available for the selected period. 2. System displays an appropriate message.                                                                                       |
| Includes                | Register / Login, Track Donation Lifecycle                                                                                                                                               |

|                             |                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| <b>Special Requirements</b> | Reports must support filtering by date, donor, NGO, and status.                                 |
| <b>Assumptions</b>          | Sufficient donation records exist in the system.                                                |
| <b>Notes and Issues</b>     | Generate Reports «includes» Track Donation Lifecycle as reports are built using lifecycle data. |

### 3.5.29 UC-29 Manage Records (Database)

| Field                    | Details                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-29                                                                                                                                    |
| <b>Use Case Name</b>     | Manage Records (Database)                                                                                                                |
| <b>Created By</b>        | Nihira Neralwar                                                                                                                          |
| <b>Last Updated By</b>   | Nihira Neralwar                                                                                                                          |
| <b>Date Created</b>      | Mar 2026                                                                                                                                 |
| <b>Date Last Updated</b> | Mar 2026                                                                                                                                 |
| <b>Actor</b>             | System Administrator                                                                                                                     |
| <b>Description</b>       | Allows the system administrator to manage system database records including archiving, deletion of obsolete data, and maintenance tasks. |
| <b>Preconditions</b>     | Administrator must be logged in with database access privileges.                                                                         |
| <b>Postconditions</b>    | Records are updated, archived, or deleted as required.                                                                                   |
| <b>Priority</b>          | Medium                                                                                                                                   |
| <b>Frequency of Use</b>  | Occasional                                                                                                                               |

|                                |                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Normal Course of Events</b> | 1. Admin opens the Database Management module. 2. Admin selects the type of operation (archive, delete, or update). 3. Admin confirms the action. 4. System executes the operation and logs it. |
| <b>Alternative Courses</b>     | 1. Admin schedules automatic archiving for old records.                                                                                                                                         |
| <b>Exceptions</b>              | 1. Record in use — deletion blocked. 2. System displays a conflict error.                                                                                                                       |
| <b>Includes</b>                | Register / Login                                                                                                                                                                                |
| <b>Special Requirements</b>    | All database changes must be logged with admin ID and timestamp.                                                                                                                                |
| <b>Assumptions</b>             | Administrator has the required privileges and backups are in place.                                                                                                                             |
| <b>Notes and Issues</b>        | This use case «extends» to send notifications if records change or errors occur.                                                                                                                |

### 3.5.30 UC-30 Real-Time Notification System

| Field                    | Details                       |
|--------------------------|-------------------------------|
| <b>Use Case ID</b>       | UC-30                         |
| <b>Use Case Name</b>     | Real-Time Notification System |
| <b>Created By</b>        | Nihira Neralwar               |
| <b>Last Updated By</b>   | Nihira Neralwar               |
| <b>Date Created</b>      | Mar 2026                      |
| <b>Date Last Updated</b> | Mar 2026                      |
| <b>Actor</b>             | System Administrator          |

|                                |                                                                                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>             | Manages the dispatch of real-time notifications to all stakeholders based on system events such as donation acceptance, task assignment, and delivery completion.                       |
| <b>Preconditions</b>           | A triggering event must occur within the system.                                                                                                                                        |
| <b>Postconditions</b>          | Notification is delivered to the appropriate user in real time.                                                                                                                         |
| <b>Priority</b>                | High                                                                                                                                                                                    |
| <b>Frequency of Use</b>        | Very Frequent                                                                                                                                                                           |
| <b>Normal Course of Events</b> | 1. A trigger event occurs such as a donation being accepted. 2. System identifies the recipients. 3. System dispatches the notification via in-app or email. 4. Notification is logged. |
| <b>Alternative Courses</b>     | 1. Admin configures custom notification rules.                                                                                                                                          |
| <b>Exceptions</b>              | 1. Notification delivery fails. 2. System retries and logs the failure.                                                                                                                 |
| <b>Includes</b>                | Register / Login                                                                                                                                                                        |
| <b>Special Requirements</b>    | Notifications must be delivered within 3 seconds of the triggering event.                                                                                                               |
| <b>Assumptions</b>             | Users have active notification preferences enabled.                                                                                                                                     |
| <b>Notes and Issues</b>        | Standalone system use case triggered by events across all modules.                                                                                                                      |

### 3.5.31 UC-31 Track Donation Lifecycle

| Field                | Details                  |
|----------------------|--------------------------|
| <b>Use Case ID</b>   | UC-31                    |
| <b>Use Case Name</b> | Track Donation Lifecycle |
| <b>Created By</b>    | Nihira Neralwar          |

|                                |                                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Last Updated By</b>         | Nihira Neralwar                                                                                                                                                        |
| <b>Date Created</b>            | Mar 2026                                                                                                                                                               |
| <b>Date Last Updated</b>       | Mar 2026                                                                                                                                                               |
| <b>Actor</b>                   | System Administrator                                                                                                                                                   |
| <b>Description</b>             | Allows the system administrator to view and track the complete lifecycle of any donation from submission through to delivery and reconciliation.                       |
| <b>Preconditions</b>           | Administrator must be logged in. Donation records must exist.                                                                                                          |
| <b>Postconditions</b>          | Full lifecycle timeline is displayed for the selected donation.                                                                                                        |
| <b>Priority</b>                | High                                                                                                                                                                   |
| <b>Frequency of Use</b>        | Frequent                                                                                                                                                               |
| <b>Normal Course of Events</b> | 1. Admin searches for a donation by ID or donor. 2. System retrieves the full lifecycle history. 3. System displays each stage with timestamps and responsible actors. |
| <b>Alternative Courses</b>     | 1. Admin filters by lifecycle stage or date.                                                                                                                           |
| <b>Exceptions</b>              | 1. Donation record not found. 2. System displays an error message.                                                                                                     |
| <b>Includes</b>                | Register / Login                                                                                                                                                       |
| <b>Special Requirements</b>    | Lifecycle data must be read-only and tamper-resistant for audit purposes.                                                                                              |
| <b>Assumptions</b>             | Donation records are continuously updated throughout the lifecycle.                                                                                                    |
| <b>Notes and Issues</b>        | This is the base use case included by Generate Reports.                                                                                                                |

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### 3.5.32 UC-32 Inspect Food Donation

| Field                   | Details                                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID             | UC-32                                                                                                                                                                                                                                                                              |
| Use Case Name           | Inspect Food Donation                                                                                                                                                                                                                                                              |
| Created By              | Nihira Neralwar                                                                                                                                                                                                                                                                    |
| Last Updated By         | Nihira Neralwar                                                                                                                                                                                                                                                                    |
| Date Created            | Mar 2026                                                                                                                                                                                                                                                                           |
| Date Last Updated       | Mar 2026                                                                                                                                                                                                                                                                           |
| Actor                   | Food Safety Inspector                                                                                                                                                                                                                                                              |
| Description             | Allows a food safety inspector to formally inspect a food donation and record the inspection outcome before the NGO accepts or rejects it.                                                                                                                                         |
| Preconditions           | Inspector must be assigned to the donation by the NGO.                                                                                                                                                                                                                             |
| Postconditions          | Inspection result is recorded and the NGO is notified to proceed with acceptance or rejection.                                                                                                                                                                                     |
| Priority                | High                                                                                                                                                                                                                                                                               |
| Frequency of Use        | Frequent                                                                                                                                                                                                                                                                           |
| Normal Course of Events | 1. Inspector receives an inspection assignment notification. 2. Inspector accesses the donation details. 3. Inspector conducts the physical and system inspection. 4. Inspector records the result as Safe, Unsafe, or Requires Review. 5. System notifies the NGO of the outcome. |
| Alternative Courses     | 1. Inspector requests additional documentation from the donor.                                                                                                                                                                                                                     |



|                             |                                                                                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Exceptions</b>           | 1. Inspector cannot access the donation details. 2. System displays an error and alerts admin.                     |
| <b>Includes</b>             | Register / Login                                                                                                   |
| <b>Special Requirements</b> | Inspection results must be logged with timestamp, inspector ID, and detailed notes.                                |
| <b>Assumptions</b>          | Inspector is assigned and the donation record is accessible.                                                       |
| <b>Notes and Issues</b>     | Shared use case included by Accept Food Donation and Reject Food Donation. Base for Flag Unsafe Food via «extend». |

### 3.5.33 UC-33 Define Food Safety Rules

| Field                    | Details                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-33                                                                                                                   |
| <b>Use Case Name</b>     | Define Food Safety Rules                                                                                                |
| <b>Created By</b>        | Nihira Neralwar                                                                                                         |
| <b>Last Updated By</b>   | Nihira Neralwar                                                                                                         |
| <b>Date Created</b>      | Mar 2026                                                                                                                |
| <b>Date Last Updated</b> | Mar 2026                                                                                                                |
| <b>Actor</b>             | Food Safety Inspector                                                                                                   |
| <b>Description</b>       | Allows a food safety inspector to define, update, and publish food safety rules that govern how donations are assessed. |
| <b>Preconditions</b>     | Inspector must be logged in with rule management privileges.                                                            |
| <b>Postconditions</b>    | Food safety rules are updated and published to the system.                                                              |

|                                |                                                                                                                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Priority</b>                | Medium                                                                                                                                                                                                  |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                                                              |
| <b>Normal Course of Events</b> | 1. Inspector navigates to the Safety Rules management section. 2. Inspector adds or updates a rule. 3. System validates and saves the rule. 4. Updated rules are published and accessible to all users. |
| <b>Alternative Courses</b>     | 1. Inspector deactivates an outdated rule.                                                                                                                                                              |
| <b>Exceptions</b>              | 1. Rule conflicts with an existing rule. 2. System alerts the inspector for review.                                                                                                                     |
| <b>Includes</b>                | Register / Login                                                                                                                                                                                        |
| <b>Special Requirements</b>    | Rules must follow regulatory food safety standards.                                                                                                                                                     |
| <b>Assumptions</b>             | Inspector has authority to define and update safety standards.                                                                                                                                          |
| <b>Notes and Issues</b>        | Standalone use case. Rules defined here govern the Inspect Food Donation use case.                                                                                                                      |

### 3.5.34 UC-34 Flag Unsafe Food

| Field                    | Details          |
|--------------------------|------------------|
| <b>Use Case ID</b>       | UC-34            |
| <b>Use Case Name</b>     | Flag Unsafe Food |
| <b>Created By</b>        | Nihira Neralwar  |
| <b>Last Updated By</b>   | Nihira Neralwar  |
| <b>Date Created</b>      | Mar 2026         |
| <b>Date Last Updated</b> | Mar 2026         |

|                                |                                                                                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>                   | Food Safety Inspector                                                                                                                                                                                              |
| <b>Description</b>             | Allows a food safety inspector to formally flag a donation as unsafe when it fails safety standards during inspection. «extends» Inspect Food Donation.                                                            |
| <b>Preconditions</b>           | Inspection must have been conducted and the food found to be unsafe.                                                                                                                                               |
| <b>Postconditions</b>          | Donation is flagged as Unsafe, the NGO is notified, and the Manage Unsafe Food workflow is triggered.                                                                                                              |
| <b>Priority</b>                | High                                                                                                                                                                                                               |
| <b>Frequency of Use</b>        | Occasional                                                                                                                                                                                                         |
| <b>Normal Course of Events</b> | 1. Inspector completes the inspection and finds the food to be unsafe. 2. Inspector selects Flag as Unsafe. 3. Inspector enters the reason and violation details. 4. System flags the donation and alerts the NGO. |
| <b>Alternative Courses</b>     | 1. Inspector recommends the donor remedy the issue and resubmit.                                                                                                                                                   |
| <b>Exceptions</b>              | 1. Flagging without a reason is not permitted. 2. System requires a description before proceeding.                                                                                                                 |
| <b>Includes</b>                | Register / Login                                                                                                                                                                                                   |
| <b>Special Requirements</b>    | Flag record must include violation type, description, and inspector ID.                                                                                                                                            |
| <b>Assumptions</b>             | Inspection was conducted and a clear safety violation was identified.                                                                                                                                              |
| <b>Notes and Issues</b>        | This use case «extends» Inspect Food Donation and is only triggered when food fails safety standards.                                                                                                              |

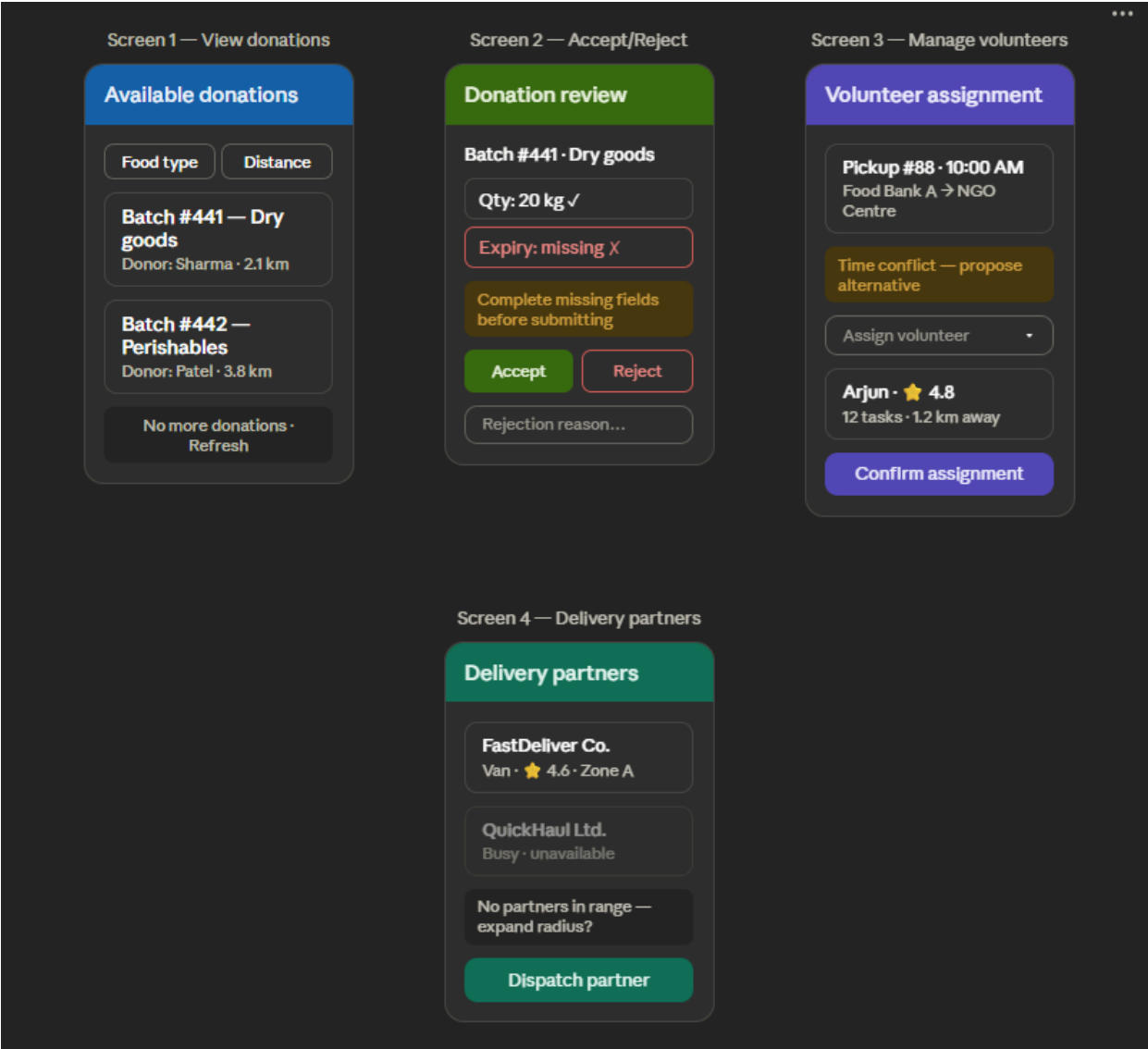
UI -

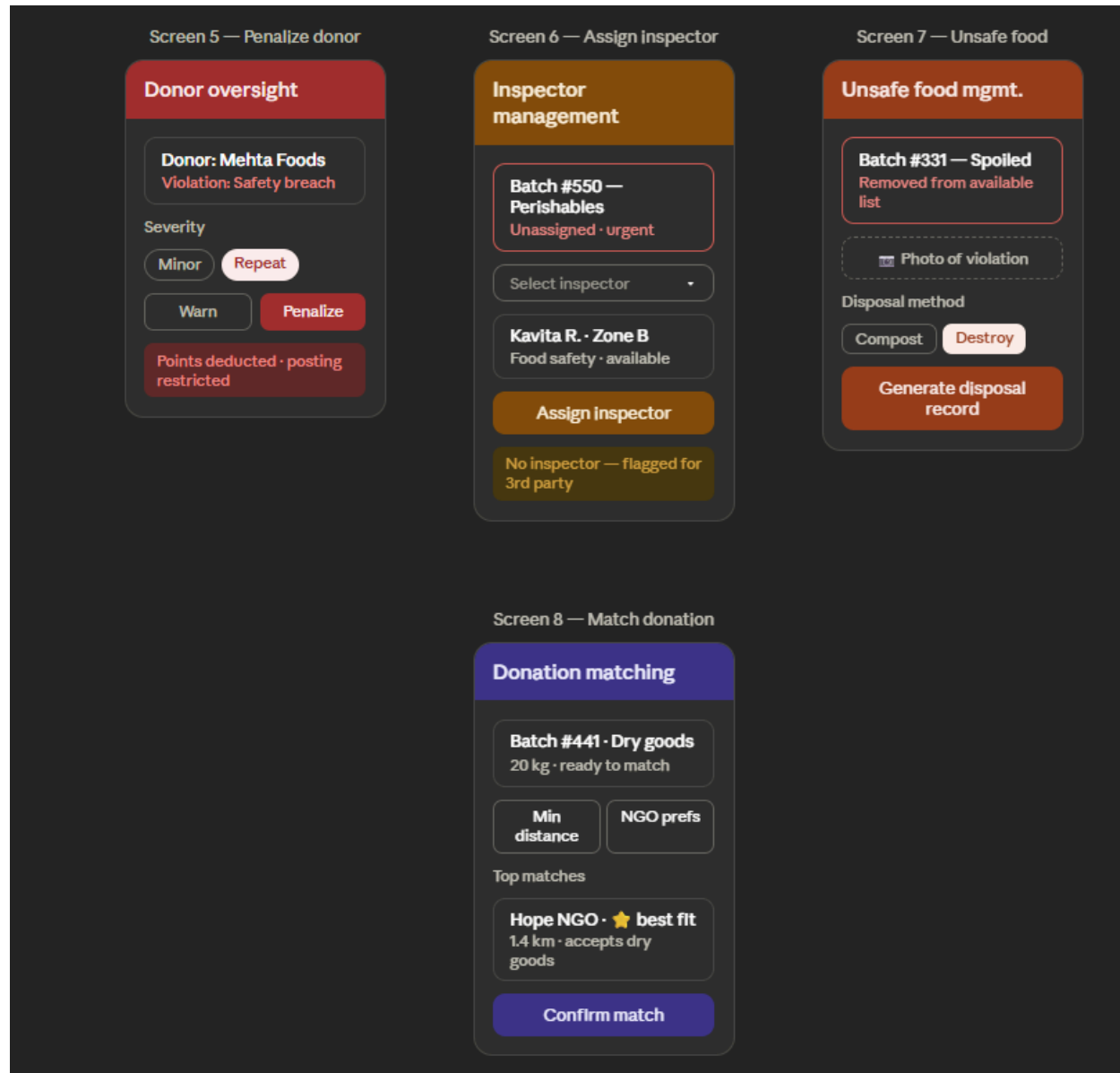
## DONOR

The image displays a four-screen user interface for a donor application, set against a dark grey background. Each screen is a rounded rectangle with a distinct header bar.

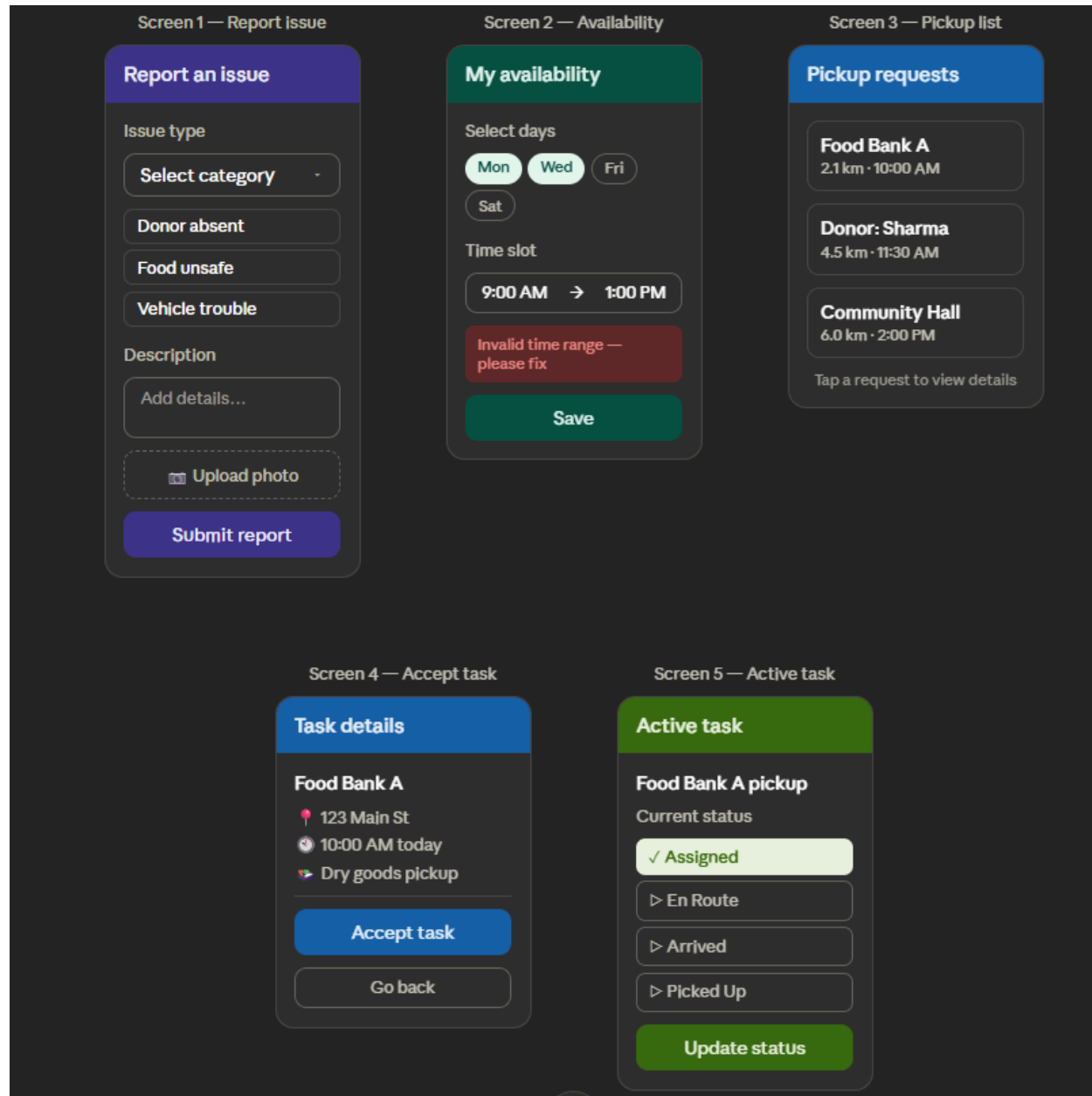
- Screen 1 — Login:** Features a purple header with the title "Donor login". It includes a "Username" section with a text input field containing the placeholder "Enter username...", and a "Password" section with a text input field containing seven dots. A red error message "Incorrect username or password" is shown below the password field. A purple "Log in" button is at the bottom, with a "Forgot password?" link underneath it.
- Screen 2 — Add donation:** Features a green header with the title "Add food donation". It includes a "Food type" dropdown menu, a "Quantity (kg)" text input field, and an "Expiry date" text input field. A red error message "Expiry date is required" is shown below the expiry date field. A green "Submit donation" button is at the bottom, with a green confirmation message "Donation added · ID #4421" below it.
- Screen 3 — Safety guidelines:** Features an orange header with the title "Food safety rules". It includes a section "Read and acknowledge all rules" with three checkboxes: "Temperature standards", "Packaging requirements", and "Expiry date policy". An orange message "All boxes must be checked" is shown below the checkboxes. An orange "Submit acknowledgment" button is at the bottom, with a green confirmation message "Verified · Add Donation unlocked" below it.
- Screen 4 — Feedback & rating:** Features a blue header with the title "Leave feedback". It includes a section "Donation #4401" with the subtext "Completed · 2 Apr". Below this is a "Star rating" section with five stars, the first four of which are filled. A text input field for "Written comments..." is below the stars. A red error message "Rating is required" is shown below the stars. A blue "Submit review" button is at the bottom, with a green confirmation message "Thanks for your feedback!" below it.

NGO





VOLUNTEER



DELIVERY PARTNER

Screen 1 — Accept delivery

Available tasks

≤ 5 km

Van

Task #1042

Pickup: Food Bank A

Drop-off: NGO Centre

Accept

Skip

Task accepted — proceed to pickup

Screen 2 — Pick up food

Pickup at donor

✓ GPS verified — within range

Scan QR / enter ID

Quantity match?

✓ Matches

Adjust qty

Reason for discrepancy...

Confirm pickup

Screen 3 — Deliver food

Delivery drop-off

NGO Centre

Confirm recipient available

Available

Contact them

Proof of delivery

🔥 Sign

📷 Photo

Submit delivery

👏 Delivery successful — points awarded

Screen 4 — Update status

Active delivery

Task #1042 · In progress

Update status

● In Transit

○ Delayed

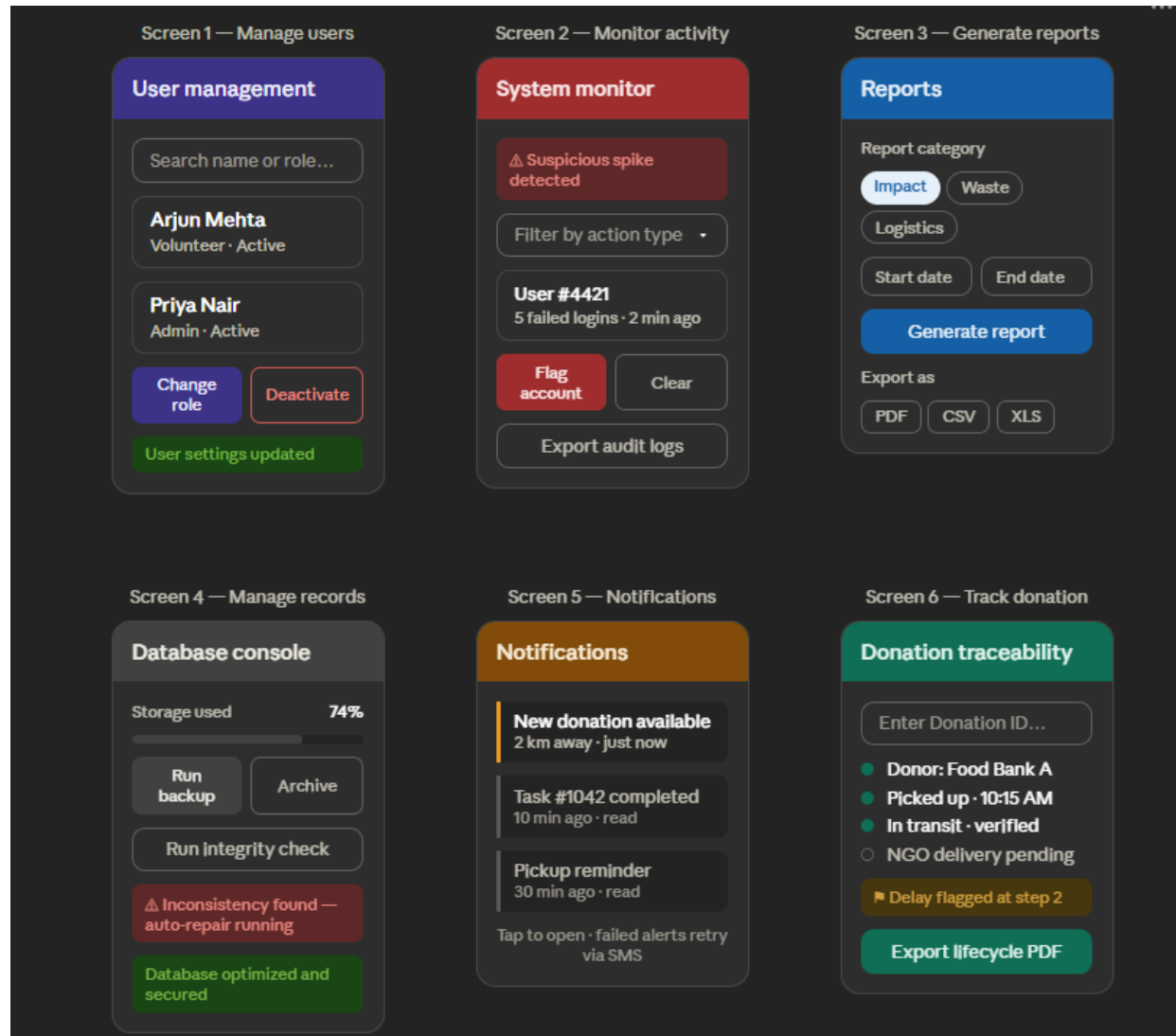
Delay reason...

Est. extra time (mins)

Submit update

SYSTEM ADMINISTRATOR





## FOOD SAFETY INSPECTOR

